Does Short Interest Predict Short - Horizon Returns?

Evidence from S&P 500 Stocks

BMF5324 - AI and Statistics

Group Members: Lo Hao Yu, Michael Steven, Vini Shrivastava, Zhang Licheng

Research Period: September 2016 - September 2026

Section 1: Executive Summary

Our call: an S&P 500 equity manager should treat high short interest as a warning, not a buy signal, and use it only as a screen on trades already planned, never as a stand-alone strategy. **Main risk**: the effect is small, about 4 to 6 bps per standard deviation over five days, it is not significant when stocks are weighted by market value, and it has weakened since 2021, so it could fade further.

FINRA publicly releases short-interest data, the number of shares sold short and not yet covered, twice a month. We examine whether these disclosures predict returns over the following five trading days. Two explanations compete: informed selling suggests that heavily shorted stocks will underperform, while covering pressure suggests that future repurchases by short sellers may push prices higher. Our original H1 predicted that higher short interest relative to float would predict higher five-day market-adjusted returns. H2 further predicted that this positive relationship would be stronger among firms with lower analyst coverage, where information asymmetry is likely greater.

To test these predictions, we use LSEG Workspace to construct a historical S&P 500 panel covering 240 short-interest releases from September 2016 to September 2026. The H1 sample contains 103,920 stock-release observations. We estimate equal-weighted Fama-MacBeth regressions with seven controls of Boehmer et al. (2022) and Newey-West standard errors using four lags. The baseline return window begins at the publication-day close, before the after-close disclosure; a one-day-delayed window assesses timing sensitivity.

- **H1 is not supported, and the evidence points the other way.** A one-standard deviation increase in short interest (2.72 percentage points of float) predicts a 6.2 basis-point lower market-adjusted return over the next five trading days ($t = -2.89$); its 95% confidence interval, −10.4 to −2.0 bps, excludes a positive average coefficient in this specification. From one release to the next the shortfall is about 11 bps ($t = -3.34$), and it does not reverse over the following two weeks. The negative association is consistent with informed selling but does not rule out covering pressure in particular stocks or episodes.
- **H2 is not supported as stated.** The $SI \times analyst$ coverage interaction is positive but insignificant ($+1.75\ bps, t = 1.22$), while a supplementary size-interaction analysis finds a significantly more negative short-interest relationship among smaller constituents ($SI \times size$: $+5.95\ bps, t = 2.94$). Coverage-based subgroup estimates point in the same direction, but the difference between low- and high-coverage firms is not established as statistically significant. The findings suggest size-related heterogeneity, with less conclusive evidence for analyst coverage; they neither confirm the original positive covering-pressure prediction nor establish information asymmetry as the cause.

Section 2: Problem Statement and Hypotheses

2.1 Research Motivation and Problem Statement

Motivation: The decision-maker is a large-cap equity manager assessing whether publicly disclosed short interest should influence investment decisions concerning heavily shorted S&P 500 stocks. Our interest in the question began with Opendoor Technologies (NASDAQ: OPEN), a U.S. residential real estate company that buys and resells homes. In July 2025, its shares rallied sharply amid renewed retail investor interest, and market commentary linked the rally partly to short covering and momentum trading (CNBC, 2025). These accounts do not establish what caused the increase, but they raise a broader question: should substantial short interest be interpreted as a warning of future underperformance, or as a potential source of buying pressure? Opendoor motivates this question, while our empirical analysis

examines whether the proposed relationship extends to a broader sample of historical S&P 500 constituents.

The conflict: High short interest may reflect negative information or overvaluation, predicting lower subsequent returns. Alternatively, concentrated short covering may create temporary buying pressure. Existing studies provide evidence for informed selling and for squeezes in specific event settings, but these findings do not establish which prediction holds around routine disclosures for S&P 500 stocks.

Timing: FINRA collects short positions on the 15th and the last business day of each month and publishes on the seventh trading day after each settlement date. A manager can act only on the publication, so the test starts there. Returns run from the close of publication date t to the close of $t + 5$, using only information public by t.

Question and contribution: Motivated by this tension, we examine whether publicly available short interest as a percentage of float predicts higher subsequent five-trading-day market-adjusted returns among historical S&P 500 constituents (H1), and whether the relationship varies with analyst coverage, our primary proxy for the information environment (H2). The study provides a focused assessment of these predictions in a large-cap sample, using publication-date observations through 10 September 2026. For portfolio managers, its purpose is to assess whether high short interest provides a useful basis for anticipating short-term gains or instead warrants closer scrutiny because of its potential negative information content. Its usefulness for investment decisions depends on the magnitude, stability and implementability of the estimated relationship, as well as its statistical significance.

2.2 H1

Claim: Across S&P 500 constituents, for FINRA publication dates from 15 September 2016 to 31 August 2026, short interest as a percentage of float, recorded at the corresponding settlement date and publicly available on the publication date, predicts a higher five-trading-day market-adjusted return, controlling for size, book-to-market, one-month and six-month lagged returns, turnover, idiosyncratic volatility and the share of zero-return days.

Null: The short interest coefficient is zero or negative.

Sink: H1 is not supported if the short-interest coefficient is not significantly positive in the primary equal-weighted specification. A significantly negative coefficient would contradict H1 and be consistent with the informed-selling interpretation.

Rationale: Outstanding short positions represent potential future repurchases. If covering demand becomes sufficiently concentrated, it may generate buying pressure and a positive relationship between publicly disclosed short interest and subsequent returns.

Test: Fama-MacBeth cross-sectional regressions (Fama and MacBeth, 1973) at each publication period, averaged over time, with Newey-West standard errors (Newey and West, 1987; four lags). The main run is equal-weighted and the value-weighted run is a robustness check. The controls follow Boehmer et al. (2022), and Section 4 gives their construction.

2.3 H2

Claim: Across the same universe and period, the H1 relationship between short interest and five-day returns is stronger for firms with lower analyst coverage, which used here as a proxy for a less-developed information environment. Coverage is the natural logarithm of one plus the number of analysts with active earnings forecasts on the publication date t, using only estimates available by t.

Null: The relationship is not stronger when coverage is lower. Equivalently, the interaction coefficient between short interest and analyst coverage is zero or positive.

Sink: The interaction is statistically indistinguishable from zero or positive, or the short interest slope is not positive among low-coverage firms.

Rational: Hong, Lim and Stein (2000) find stronger momentum among stocks with lower analyst coverage after controlling for size, particularly among past losers. Their findings are consistent with slower information diffusion, motivating analyst coverage as an imperfect proxy for the information environment. H2 extends H1's covering-pressure mechanism. We conjecture that this pressure persists longer in less-covered firms, where information about fundamental value is less widely disseminated, strengthening the positive relationship between short interest and subsequent five-day returns.

Test: Add analyst coverage and short interest × coverage to the H1 Fama-MacBeth run, with the same controls. H2 predicts a negative interaction. Report the implied short-interest slope at low and high coverage, and re-estimate the H1 regression separately for the lowest and highest coverage terciles. We additionally examine an exploratory interaction between short interest and firm size.

Section 3: Relevant Literature

3.1 Short Interest and Informed Selling

The informed-selling literature provides a foundation for expecting a negative relationship between short interest and subsequent returns. Desai et al. (2002) find that heavily shorted Nasdaq stocks earn negative abnormal returns, consistent with short sellers identifying overvalued securities. Asquith, Pathak, and Ritter (2005) extend this perspective by examining short interest jointly with institutional ownership. Stocks combining high short interest with low institutional ownership face both substantial shorting demand and potentially limited lending supply. Their underperformance is particularly pronounced in equal-weighted portfolios, while value-weighted results are weaker. These findings suggest that the predictive content of short interest depends on stock characteristics and portfolio construction. They motivate testing the relationship within the S&P 500 rather than assuming that results from broader or more constrained samples transfer unchanged to large-cap stocks.

3.2 Short Covering and Temporary Price Pressure

Short covering provides a competing mechanism through which heavily shorted stocks may experience positive returns. Closing short positions requires purchasing shares, and concentrated purchases can amplify price increases. Stice-Lawrence, Wong, and Zhao (2025) examine public short-selling attacks and document evidence consistent with short squeezes in a subset of events, including increased covering and subsequent reversals of price gains. This supports the relevance of covering pressure under particular conditions, but does not establish an unconditional positive relationship between short interest and future returns. Price increases can also attract new short selling: Diether, Lee, and Werner (2009) find that short sellers increase activity following positive returns and that this activity predicts lower subsequent returns. Their evidence cautions against interpreting short selling around rising prices as proof of a squeeze. These findings leave open whether covering-related price pressure translates into a positive predictive relationship around routine short-interest disclosures.

3.3 Alternative Measures of Short Interest

Different short-selling measures capture different economic features. Boehmer et al. (2022) compare multiple measures across 38 countries and find that days-to-cover and utilization have particularly robust predictive power, with substantial differences across firms and markets. Our SI/float measure captures outstanding short positions relative to publicly tradable shares, but does not directly measure borrowing availability or the trading volume available for covering. Hanauer, Lesnevski, and Smajlbegovic (2023)

further show that a measure extracting the surprise component of short interest negatively predicts returns. Their measure differs from both the short-interest level and a simple change in that level. These distinctions motivate interpreting our results specifically as evidence about SI/float. Boehmer et al. (2022) also inform our choice of controls for size, valuation, liquidity, volatility, and past returns; the implemented definitions are presented in Section 4.

3.4 Analyst Coverage and Information Diffusion

The information-diffusion literature provides a basis for examining analyst coverage in return predictability. Hong and Stein (1999) develop a model in which information spreads gradually across investors, causing prices to underreact initially and adjust over time. Hong, Lim, and Stein (2000) provide related empirical evidence: after accounting for firm size, momentum profits are greater among stocks with lower analyst coverage, with the coverage effect particularly pronounced among past losers. Their findings are consistent with slower incorporation of negative information into the prices of less-covered firms. Analyst coverage therefore provides an observable proxy for differences in firms' information environments, although it is not a direct measure of information asymmetry.

Research on short interest also suggests that the information environment matters for return predictability. Hanauer, Lesnevski, and Smajlbegovic (2023) find that short-interest surprises negatively predict subsequent returns, with stronger predictability among stocks with greater information uncertainty. Although information uncertainty differs from analyst coverage, this evidence complements the information-diffusion literature by showing that the predictive content of short-selling signals varies across information environments. Together, these studies motivate examining whether the relationship between short interest and subsequent returns differs across levels of analyst coverage.

3.5 Research Positioning

Our study applies these competing arguments to historical S&P 500 constituents, using publicly available short-interest observations to predict returns over the following five trading days. The contribution is a focused assessment of the signal in this sample and information-timing framework, rather than a claim that large-cap stocks or five-day predictability have not previously been studied. We examine whether the positive relationship proposed in H1 appears and whether analyst coverage or size moderates it. A negative estimate would challenge our original prediction and be consistent with informed selling, while leaving the underlying causal mechanism unresolved.

Section 4: Data and Methodology

4.1 Universe and sample

Our sample covers 240 FINRA short-interest publication dates from 26 September 2016 to 10 September 2026, corresponding to settlement dates from 15 September 2016 to 31 August 2026. We use historical S&P 500 membership from LSEG Workspace, including stocks only when they belonged to the index. This reduces survivorship bias in constructing the initial sample, although missing data for delisted firms remain a limitation.

After matching historical identifiers, retaining one share class per company, and excluding unreliable records, the panel contains 119,879 stock-release observations across 688 securities. Market, short-interest, accounting, and analyst data come from LSEG Workspace, while factor returns are obtained from the Kenneth French Data Library.

LSEG lacks short-interest histories for 101 subsequently delisted constituents, representing 7.7% of observations. The final H1 regression sample contains 103,920 observations, averaging 433 stocks across 240 releases, while H2 averages 429 stocks per release.

4.2 Event timing

FINRA collects short positions twice each month, the 15th and the last business day and publishes them seven trading days later. We rebuild this schedule and verify it against FINRA's published calendar and Nasdaq's dated releases. The lag from settlement to publication is 9-13 calendar days. Each observation is indexed by its publication date t. Short interest is measured at settlement but enters the model only at t, when it becomes public. All other predictors use only information available by the close of t:

- Membership, market capitalisation and analyst coverage are measured on day t;
- Return, turnover and volatility windows end at the close of t;
- Book equity is matched using its original announcement date where available. When the announcement date is missing, we apply a 90-day lag from the fiscal quarter-end as an approximation of when the information became public.

The dependent variable starts where the predictors end: the market-adjusted return from the close of t to the close of $t + 5$. Each FINRA publication date forms one Fama-MacBeth cross-section (240 in total). We also examine alternative return windows and a one-day delay to assess the sensitivity of our results to return timing, as described in Section 4.5.

4.3 Variables

Appendix Table A defines the variables. We follow Boehmer et al. (2022) in selecting seven firm-level controls to account for established return-related characteristics, including size, valuation, momentum, volatility, and liquidity. This helps distinguish the predictive relationship of short interest from other firm characteristics that may also affect future returns. Short interest, float and volume are converted to the same share units on each date using LSEG's split and spin-off adjustment factors, separately for each share class. Book-to-market is set to missing when book equity is negative (6,629 observations; a robustness test keeps these firms with a dummy). Continuous variables, except analyst coverage and the share of zero-return days, are winsorised at the 1st and 99th percentiles within each release date. Regressors are standardized within each date, so coefficients are basis points of return per one-standard-deviation change.

4.4 Estimation

We use the Fama-MacBeth two-step procedure to examine whether publicly disclosed short interest predicts future stock returns. For each of the 240 FINRA publication dates, we run a cross-sectional regression across S&P 500 constituents, allowing the effects of firm characteristics to vary over time rather than imposing constant coefficients as in pooled regression.

The dependent variable is the subsequent five-trading-day stock return minus the S&P 500 return over the same period, measured from the publication-day close. All continuous predictors are standardized within each release date for comparability. For H2, the interaction term is the product of standardized short interest and analyst coverage, without further standardization. Coefficients on the standardized predictors are expressed in basis points per one-standard-deviation increase, while interaction effects are interpreted conditionally.

The baseline specification for H1 is:

$$R_{i,t+5}^{adj} = \alpha_t + \beta_t SI_{i,t} + \gamma_t^T X_{i,t} + \varepsilon_{i,t}$$

where $SI_{i,t}$ denotes short interest as a percentage of free float and $X_{i,t}$ contains seven controls: log market capitalization, book-to-market ratio, one-month lagged return, return over months -6 to -2, turnover, idiosyncratic volatility, and the proportion of zero-return days.

To test H2, we augment the baseline specification with analyst coverage and its interaction with short interest:

$$R_{i,t+5}^{adj} = \alpha_t + \beta_t SI_{i,t} + \delta_t Coverage_{i,t} + \theta_t\left(SI_{i,t} \times Coverage_{i,t}\right) + \gamma_t^T X_{i,t} + \varepsilon_{i,t}$$

Analyst coverage is measured as the natural logarithm of one plus the number of covering analysts. Both short interest and analyst coverage are standardized before constructing the interaction term. We additionally estimate an alternative interaction specification using log market capitalization in place of analyst coverage.

Each cross-sectional regression is estimated by ordinary least squares, assigning equal weight to each eligible stock. We use equal weighting to estimate the relationship for a typical S&P 500 constituent without allowing the largest firms to dominate the regression. Market-cap-weighted regressions are conducted as a robustness check. The Fama-MacBeth coefficient estimate is the time-series average of the corresponding cross-sectional coefficients:

$$\bar{\beta} = \frac{1}{T}\sum_{t=1}^{T} \hat{\beta}_t$$

Since FINRA releases occur roughly twice a month, the estimated cross-sectional coefficients may be correlated across periods. We therefore use Newey-West heteroskedasticity and autocorrelation - consistent standard errors with four lags and test alternative lag lengths to check whether our results are sensitive to this choice.

4.5 Robustness

We conduct several robustness checks to assess whether the estimated relationship depends on portfolio weighting, variable definitions, return horizons, or sample composition. The full results are reported in Appendix Table D.

First, we re-estimate the baseline and analyst-coverage interaction models using weighted least squares, with market capitalization as the observation weight. This assesses whether the results persist when larger firms receive greater influence. We also replace short interest relative to free float with short interest relative to shares outstanding and repeat the baseline regression using the uncleaned panel.

Second, we examine two consecutive release-to-release return windows, from t to $t+1$ and from $t+1$ to $t+2$, using short interest observed at t to assess whether its predictive relationship persists beyond the first release cycle. We also shift the five-day return window forward by one trading day to assess sensitivity to the timing of return measurement.

Third, we examine changes in short interest between consecutive reporting periods. We add the two-week percentage change in short interest to the baseline model and separately include its interaction with the short-interest level. Both specifications are estimated over the five-day and biweekly return horizons.

Finally, we assess sample sensitivity by retaining firms with non-positive book equity through a separate indicator, estimating the models over the 2016-2020 and 2021-2026 subperiods, and comparing the characteristics of firms with and without available short-interest data. We also estimate the baseline model separately for stocks in the lowest and highest analyst-coverage terciles, formed within each publication date, to examine heterogeneity in the short-interest relationship.

Section 5: Key Findings

5.1 Descriptives

Exhibit 1 summarizes the 103,920 stock-release observations used in the H1 regressions, covering 240 FINRA publication dates. Short interest averages 2.95% of free float, with a median of 2.04% and a 90th percentile of 6.09%. The average five-day market-adjusted return is −0.08%, with a standard deviation of 3.67%. Short interest is negatively correlated with firm size (−0.46) and analyst coverage (−0.10), while firm size and analyst coverage are positively correlated (+0.46). These correlations suggest that firm characteristics may be important when examining the relationship between short interest and returns.

Exhibit 1: Descriptive statistics, H1 regression sample

Variable	Mean	SD	P10	Median	P90
$Short\ interest\ /\ float$ (%)	2.95	2.72	0.93	2.04	6.09
$5-day\ market-adjusted\ return$ (%)	-0.08	3.67	-4.20	-0.07	4.04
$Market\ cap$ ($\$\ billion$)	80.3	236.3	10.3	29.0	155.8
$Book-to-market$	0.37	0.31	0.07	0.29	0.76
$Turnover$ (% $of\ shares\ per\ day$)	0.85	0.59	0.39	0.68	1.45
$Zero-return\ days$ (% $of\ last\ 21\ days$)	0.44	1.50	0.00	0.00	0.00
$Idiosyncratic\ volatility$ (% $per\ day$)	1.46	0.66	0.83	1.30	2.29
$Analysts\ (number)$	18.3	7.8	9	18	28
$Return, past\ month$ (%)	1.00	8.69	−8.92	1.04	10.92
$Return, months\ -6\ to\ -2$ (%)	4.93	19.61	−17.52	3.96	27.56

Notes: Pooled over the 103,920 stock-release observations in the H1 regressions (240 releases, 570 firms, 433 stocks per release on average). Market capitalization and analyst counts are reported in their original units, and zero-return days are shown without winsorization. Other continuous variables are winsorized at the 1st and 99th percentiles within each release. P10 and P90 are the 10th and 90th percentiles.

5.2 H1 result

Exhibit 2 reports the Fama-MacBeth regression results for H1. We expected higher short interest to predict higher subsequent five-day market-adjusted returns because of potential short-covering pressure. However, the results show the opposite. In Model (1), the short-interest coefficient is negative but insignificant ($-3.99\ bps, t = -1.38$). After adding the seven firm-level controls in Model (2), the coefficient becomes more negative and statistically significant ($-6.22\ bps, t = -2.89$). This means that a one-standard-deviation increase in short interest is associated with 6.22 bps lower subsequent five-day market-adjusted returns, holding the other variables constant. H1 is therefore not supported.

The negative relationship is consistent with the informed-selling literature. Boehmer et al. (2022) find that short-selling measures can predict lower future returns. Hanauer et al. (2023) further show that unexpected short interest is negatively related to subsequent returns. Unlike their study, which measures short-interest surprises relative to a stock's historical mean and volatility, our main model uses the level of short interest. This distinction matters because high short interest does not necessarily indicate new short-selling activity or imminent covering.

Our model also cannot directly identify whether short covering drives stock returns, since it measures outstanding short positions rather than actual covering trades. Stice-Lawrence et al. (2025) show that short squeezes arise under specific conditions following activist short-selling attacks, rather than simply from high short interest. Our negative H1 result is therefore more consistent with informed selling than with the covering-pressure prediction. However, it does not rule out the possibility that covering pressure affects stock prices under particular market conditions.

Exhibit 2: Fama-MacBeth regressions of the next 5-day market-adjusted return (bps per 1SD)

	(1) SI only	(2) H1	(3) H2: coverage	(4) H2: size	(5) low-coverage third	(6) high-coverage third
$SI\ /\ float$	−3.99	−6.22***	−5.04**	−0.89	−8.83***	−3.71
	(−1.38)	(−2.89)	(−2.36)	(−0.34)	(−2.80)	(−0.99)
$Log\ market\ cap$		3.48	2.88	6.75**	0.38	6.43
		(1.38)	(1.15)	(2.20)	(0.16)	(1.63)
$Book-to-market$		−2.39	−2.34	−2.08	−1.39	−3.62
		(−0.70)	(−0.69)	(−0.62)	(−0.42)	(−0.87)
$Return, past\ month$		−3.30	−3.24	−3.91	−3.29	−3.55
		(−0.80)	(−0.78)	(−0.97)	(−0.81)	(−0.79)
$Return, months-6\ to-2$		−1.88	−1.71	−2.43	−0.96	−1.46
		(−0.42)	(−0.39)	(−0.54)	(−0.21)	(−0.26)
$Turnover$		4.13	3.24	4.11	2.12	4.22
		(1.20)	(0.95)	(1.18)	(0.52)	(0.91)
$Idiosyncratic\ volatility$		1.87	1.54	1.75	1.62	6.06
		(0.63)	(0.53)	(0.59)	(0.46)	(1.50)
$zero-return\ days$		0.27	0.40	0.40	1.29	0.96
		(0.27)	(0.39)	(0.39)	(0.71)	(0.49)
$Analyst\ coverage, ln(1+N)$			2.31			
			(1.01)			
$SI\ \times\ analyst\ coverage$			1.75			
			(1.22)			
$SI\ \times\ size$				5.95***		
				(2.94)		
$Intercept$	−7.52	−7.93	−7.55	−5.05	−11.51	−3.11
	(−1.28)	(−1.34)	(−1.29)	(−0.89)	(−1.58)	(−0.62)
$Releases$	240	240	240	240	240	240
$Avg.stocks\ per\ release$	461	433	429	433	144	140
$Avg.adjusted\ R^2$	0.018	0.135	0.142	0.137	0.135	0.152

*Notes: Each release is one cross-section; regressors are standardized within each release, and interactions are products of the standardized variables. Newey-West t-statistics (4 lags) in parentheses; *, **, *** = significant at 10%, 5%, 1%. The last two columns re-estimate model (2) within the bottom and top thirds of analyst coverage, formed at each release.*

5.3 Standard-error comparison

Table B in the appendix compares the statistical significance of the H1 coefficient across different inference methods. The negative short-interest coefficient remains significant using conventional Fama-MacBeth standard errors ($t = -2.30$) and Newey–West standard errors with four lags ($t = -2.89$). The Newey-West adjustment produces a smaller standard error because the period-by-period coefficients are negatively autocorrelated.

However, the result is not significant in a pooled regression with fixed control-variable coefficients and date-clustered standard errors ($t = -1.38$). When the pooled model allows control-variable coefficients to vary by date, the t-statistic returns to −2.30. This distinction matters because the relationship between firm characteristics and returns may change across market conditions. Overall, the negative H1 result is

supported by the main Fama–MacBeth specification, but its statistical significance is sensitive to the model specification and the treatment of dependence in the data.

5.4 H2 result

H2 predicted that the positive short-interest relationship would be stronger among firms with greater information asymmetry. Exhibit 2 shows that the SI × Analyst Coverage coefficient is positive but insignificant ($1.75\ bps, t = 1.22$). In contrast, the SI × Size interaction is positive and significant (5.95 bps, t = 2.94), indicating that the negative relationship found in H1 is stronger among smaller firms. The coverage tercile results point in the same direction, with SI coefficients of −8.83 bps for low-coverage firms and −3.71 bps for high-coverage firms, although the difference is not established as statistically significant.

One possible explanation is that negative information spreads more slowly among smaller and less-covered firms. Hong, Lim, and Stein (2000) provide evidence of slower information diffusion in these firms, particularly for negative information. Hanauer et al. (2023) also find that short-interest surprises have stronger negative return predictability when information uncertainty is higher. Although information uncertainty differs from information asymmetry, both studies suggest that a firm's information environment may affect how quickly negative information is reflected in stock prices.

These results suggest exploratory size-related heterogeneity, but do not support the original positive H2 or establish information asymmetry as the cause.

5.5 Slope stability

Table C in the appendix groups the 240 release-by-release short-interest slopes from model (2) by year. Single slopes are noisy: their standard deviation is 42 bps, nearly seven times the mean, and only 59.6% of them are negative. The mean is negative in 9 of the 11 calendar years and strongest in 2019 ($-19.22\ bps, t = -4.10$), but positive in 2023 and in 2026 so far. The negative relationship is weaker in the later subperiod, falling from −10.64 bps ($t = -3.05$) during 2016–2020 to −2.89 bps ($t = -1.16$) during 2021–2026. A rolling one-year (24-release) mean has ranged from −21.7 to +6.1 bps. When the 240 releases are split into two halves of 120, the difference is −6.89 bps ($t = -1.64$), so the decline is not statistically significant.

Section 6: Recommendation with Limitations

6.1 Recommendation

Our findings suggest that short interest is more useful as a warning signal than as a reason to expect prices to rise due to short covering. We recommend that managers of S&P 500 equity portfolios use publicly disclosed short interest as an additional input in their investment decisions, rather than as a stand-alone trading strategy.

First, investors should not expect heavily shorted stocks to rise simply because short sellers may eventually cover their positions. Our H1 results show that higher short interest is associated with lower subsequent five-day market-adjusted returns, contrary to our original prediction. This does not rule out short covering, but high short interest alone is not enough to predict when covering will occur or whether it will push prices higher.

Second, managers could use high short interest to flag stocks for further review, particularly among smaller S&P 500 constituents where the negative relationship is stronger. They should examine the company's fundamentals, possible negative information not yet reflected in prices, and whether the original investment thesis still holds. Recent sharp price increases and changes in short interest,

especially declines, should also be reviewed as possible signs of covering activity. However, these indicators should not be treated as proof of a short squeeze or used to trigger automatic trades.

Finally, the signal should be monitored rather than assumed to remain effective. Its predictive relationship is weaker during 2021–2026 and is not statistically significant under value weighting. Without a portfolio backtest and transaction-cost analysis, we cannot conclude that a strategy based solely on short interest would be profitable. Its most practical use is therefore as a supplementary screening tool within existing portfolio reviews.

6.2 Limitations

Sample selection and data quality. LSEG lacks short-interest history for 101 delisted S&P 500 constituents, representing 7.7% of observations, with missing data concentrated in 2016-2020. These firms tend to be smaller and have higher turnover and idiosyncratic volatility. Although their five-day returns are not significantly different from those of included firms ($t = -0.62$), their absence may affect our estimates, particularly because the negative short-interest relationship is stronger among smaller stocks. In addition, our data come primarily from one vendor, and errors may remain despite adjustments for corporate actions. For book equity observations without an original announcement date, the 90-day reporting lag may not fully eliminate look-ahead bias. Returns also exclude dividends, which may introduce measurement error.

Model and inference sensitivity. The negative H1 coefficient is significant under our main Fama-MacBeth specification, but not under pooled OLS with fixed control coefficients and date-clustered standard errors ($t = -1.38$). This suggests that the statistical evidence depends partly on the estimation framework. We also examine multiple specifications, and findings such as the size interaction and subperiod differences should be treated as exploratory rather than definitive.

Economic interpretation and scope. Our model uses outstanding short interest rather than actual covering transactions. Without direct covering or securities-lending data, such as borrowing fees and utilization, we cannot establish whether short covering or informed selling drives the observed returns. The sample is also limited to S&P 500 constituents, so the findings may not apply to smaller or less liquid stocks. We do not directly test whether the relationship changes with market uncertainty.

Investment implementation. Our regressions estimate predictive relationships rather than returns from an executable trading strategy. We do not model transaction costs, borrowing fees, bid-ask spreads or market impact. Therefore, even where the short-interest coefficient is statistically significant, we cannot conclude that trading on this signal would generate positive net returns.

6.3 What would change the answer

Three types of evidence could change our conclusions. First, if the short-interest coefficient becomes consistently positive and statistically significant in future FINRA releases, we would reconsider using high short interest as a negative screening signal. However, a positive relationship alone would not prove that short covering drives returns.

Second, more direct data on covering activity and securities lending conditions could help us test whether high short interest predicts positive returns under specific squeeze conditions. This would allow us to distinguish potential covering pressure from the negative information associated with short positions, which our current model cannot directly identify.

Finally, we would reconsider using short interest as a more active investment signal if its negative relationship with future returns remains significant in new data and under value weighting. A realistic portfolio backtest would also need to show that the strategy remains profitable after trading costs.

Appendix

Table A. Variable definitions

Variable	Role	Definition	Timing / source
Future 5-day market-adjusted return	Dependent	Stock's cumulative return over the next 5 trading days minus the S&P 500 return over the same days	Close of *t* to close of *t*+5; LSEG
Short interest / float (SI)	Main independent	Shares sold short ÷ free-float shares, both at the settlement date, in the same share units	Settlement date, public at *t*; LSEG (FINRA data)
Log market cap	Control	ln(company market capitalisation)	Day *t*; LSEG
Book-to-market	Control	Latest announced common equity ÷ market capitalisation	Equity public by *t*; LSEG
LagRet1m	Control	Cumulative return over the 21 trading days ending at *t*	LSEG
LagRet6m	Control	Cumulative return from 126 to 21 trading days before *t* (skips the latest month)	LSEG
Turnover	Control	Average daily volume ÷ shares outstanding, last 21 trading days	LSEG (consolidated volume)
IdioVOL	Control	Standard deviation of residuals from a Fama-French 3-factor regression of daily excess returns, last 63 trading days	LSEG; K. French
PctZero	Control	Share of zero-return days in the last 21 trading days	LSEG
Analyst coverage	H2 conditioning	ln(1 + number of analysts with FY1 EPS estimates)	Day *t*; LSEG I/B/E/S
SI × analyst coverage	H2 test	Product of standardised SI and analyst coverage	Constructed
SI × firm size	H2 robustness	Product of standardised SI and log market cap	Constructed
ΔSI	Exploratory	Percentage change in SI/float since the previous release	*t* vs *t*−1
Bi-weekly return (*t*+1, *t*+2)	Robustness	Market-adjusted return from release *t* to release *t*+1 (and *t*+1 to *t*+2)	LSEG

Table B. Standard errors for the short-interest coefficient in model (2)

Method	Coefficient	Std. error	t-statistic
Fama-MacBeth, no lag correction	−6.22	2.70	−2.30
Fama-MacBeth, Newey-West 4 lags (main)	−6.22	2.15	−2.89

Method	Coefficient	Std. error	t-statistic
Fama-MacBeth, Newey-West 8 lags	−6.22	2.06	−3.01
Fama-MacBeth, Newey-West 12 lags	−6.22	2.05	−3.03
Pooled OLS, date effects, conventional SE	−4.43	1.45	−3.06
Pooled OLS, date effects, clustered by firm	−4.43	1.92	−2.31
Pooled OLS, date effects, clustered by date	−4.43	3.22	−1.38
Pooled OLS, control slopes vary by date, clustered by date	−6.08	2.65	−2.30

Notes: Coefficients in bps of five-day return per 1 SD of SI / float. The pooled regressions use the same within-release standardized regressors and returns demeaned by release (date fixed effects) on the 103,920 observations. The last row lets the seven control coefficients differ by release, as Fama-MacBeth does. First-order autocorrelation of the 240 Fama-MacBeth slopes: −0.16.

Table C. Short-interest slope by year, model (2)

Period	Releases	Mean slope (bps)	t-statistic	% of releases negative
2016 (from Sep)	7	−2.13	−0.27	57.1
2017	24	−13.82	−1.63	66.7
2018	24	−3.97	−0.80	45.8
2019	24	−19.22	−4.10	75.0
2020	24	−8.02	−0.99	66.7
2021	24	−2.73	−0.47	54.2
2022	24	−9.56	−1.85	58.3
2023	24	+5.26	0.82	50.0
2024	24	−5.29	−0.94	62.5
2025	24	−7.27	−1.36	66.7
2026 (to Sep)	17	+4.38	1.15	47.1
2016–2020	103	−10.64	−3.05	63.1
2021–2026	137	−2.89	−1.16	56.9
All releases	240	−6.22	−2.89	59.6

Notes: Mean of the release-by-release SI / float coefficients from model (2); Newey-West t-statistics (4 lags) within each period. Years are publication years. Standard deviation of the 240 slopes: 42.0 bps.

Table D. Robustness: short-interest coefficient across specifications (bps per 1 SD)

Specification	SI / float	Other term	Releases
Panel A. H1, model (2)			
Baseline: 5-day return, equal-weighted	−6.22 (−2.89)***		240
Value-weighted (WLS by market cap)	−4.80 (−1.35)		240
Short interest / shares outstanding	−6.56 (−2.80)***		240
Raw panel (no cleaning or winsorizing)	−11.02 (−1.72)*		240
Keep negative book equity (with dummy)	−5.94 (−2.82)***		240
5-day window starts one day later	−4.32 (−1.70)*		240

Specification	SI / float	Other term	Releases
Panel A. H1, model (2)			
Bi-weekly return, release *t* to *t*+1	−10.83 (−3.34)***		239
Bi-weekly return, release *t*+1 to *t*+2	−11.09 (−3.47)***		238
Add ΔSI, 5-day return	−6.30 (−2.94)***	ΔSI: −2.60 (−1.78)*	239
Add ΔSI and SI × ΔSI, 5-day return	−6.77 (−3.12)***	SI × ΔSI: 2.29 (1.37)	239
Add ΔSI, bi-weekly return	−11.17 (−3.43)***	ΔSI: −3.10 (−1.53)	238
Releases in 2016–2020	−10.64 (−3.05)***		103
Releases in 2021–2026	−2.89 (−1.16)		137
Panel B. H2, model (3); other term = SI × analyst coverage			
Baseline	−5.04 (−2.36)**	1.75 (1.22)	240
Value-weighted (WLS by market cap)	−3.27 (−0.95)	3.79 (1.34)	240
Bi-weekly return, release *t* to *t*+1	−9.48 (−2.96)***	1.22 (0.62)	239
Releases in 2016–2020	−9.43 (−2.90)***	3.24 (1.54)	103
Releases in 2021–2026	−1.75 (−0.66)	0.63 (0.33)	137

*Note: Each row changes one element of the baseline. Newey-West t-statistics (4 lags) in parentheses; *, **, *** = significant at 10%, 5%, 1%. ΔSI is the % change in SI / float since the previous release. Bi-weekly returns run from the close of one release day to the close of the next.*

Table E. Rival mechanisms and the sign each predicts

Mechanism	Source	H1 sign	H2 interaction sign (high asymmetry)
Covering pressure (our thesis)	Stice-Lawrence, Wong and Zhao (2025), event setting after activist short attacks	Positive	Positive, if low-information firms also cost more to cover
Informed selling	Desai et al. (2002); Asquith, Pathak and Ritter (2005); Boehmer, Jones and Zhang (2008)	Negative	Negative. Hanauer, Lesnevski and Smajlbegovic (2023) find stronger predictability under high information uncertainty
Shorting against overreaction	Diether, Lee and Werner (2009)	Negative	No direct prediction

Note: Signs in the last column are for the asymmetry direction, so a positive entry corresponds to a negative coefficient on coverage.

Reference

[1] Asquith, P., Pathak, P. A., & Ritter, J. R. (2005). Short interest, institutional ownership, and stock returns. Journal of Financial Economics, 78(2), 243–276.

[2] Boehmer, E., Huszár, Z. R., Wang, Y., Zhang, X., & Zhang, X. (2022). Can shorts predict returns? A global perspective. Review of Financial Studies, 35(5), 2428–2463.

[3] Boehmer, E., Jones, C. M., & Zhang, X. (2008). Which shorts are informed? Journal of Finance, 63(2), 491–527.

[4] CNBC. (2025, July 22). Opendoor shares turn negative as hyped real estate stock fails to sustain meme rally. https://www.cnbc.com/2025/07/22/opendoor-leads-meme-stock-redux-on-wall-street-with-shares-tripling-in-one-week.html

[5] Desai, H., Ramesh, K., Thiagarajan, S. R., & Balachandran, B. V. (2002). An investigation of the informational role of short interest in the Nasdaq market. Journal of Finance, 57(5), 2263–2287.

[6] Diether, K. B., Lee, K.-H., & Werner, I. M. (2009). Short-sale strategies and return predictability. Review of Financial Studies, 22(2), 575–607.

[7] Fama, E. F., & French, K. R. (1993). Common risk factors in the returns on stocks and bonds. Journal of Financial Economics, 33(1), 3–56.

[8] Fama, E. F., & MacBeth, J. D. (1973). Risk, return, and equilibrium: Empirical tests. Journal of Political Economy, 81(3), 607–636.

[9] Hanauer, M. X., Lesnevski, P., & Smajlbegovic, E. (2023). Surprise in short interest. Journal of Financial Markets, 65, 100841.

[10] Hanson, S. G., & Sunderam, A. (2014). The growth and limits of arbitrage: Evidence from short interest. Review of Financial Studies, 27(4), 1238–1286.

[11] Hong, H., Lim, T., & Stein, J. C. (2000). Bad news travels slowly: Size, analyst coverage, and the profitability of momentum strategies. Journal of Finance, 55(1), 265–295.

[12] Hong, H., & Stein, J. C. (1999). A unified theory of underreaction, momentum trading, and overreaction in asset markets. Journal of Finance, 54(6), 2143–2184.

[13] Nasdaq. (n.d.). Short interest. Nasdaq Trader. Retrieved October 22, 2026, from https://nasdaqtrader.com/asp/short_interest.asp

[14] Newey, W. K., & West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. Econometrica, 55(3), 703–708.

[15] Senchack, A. J., & Starks, L. T. (1993). Short-sale restrictions and market reaction to short-interest announcements. Journal of Financial and Quantitative Analysis, 28(2), 177–194.

[16] Stice-Lawrence, L., Wong, Y. T. F., & Zhao, W. (2025). Short squeezes after short-selling attacks. Journal of Accounting Research, 63(3), 1187–1236.

AI Collaboration Appendix

AI tool: Claude and Chat GPT, used through Claude Code, 17-25 October 2026. Prompts are quoted as typed. The three exchanges come from the three ways we used it most: explaining our methods to us simply, extracting the data, and checking the report against the course brief.

Exchange 1. Changed Our Analytical Direction (12 October 2026)

Prompt: Following our professor's suggestion, we want to replace our original H2, which predicted a stronger positive short-interest relationship after positive preceding five-day returns, with a hypothesis about information asymmetry. How can we use analyst coverage to develop this conditional prediction and test it?

Response (one sentence): ChatGPT helped translate the professor's suggestion into an analyst-coverage interaction test, drawing on information-diffusion research while distinguishing its findings from our conjecture that covering-related positive price pressure would be stronger among less-covered firms.

What it changed: We shifted H2 from recent price performance to the firm's information environment, using analyst coverage as a proxy for information asymmetry. We replaced the interaction between short interest and a positive-prior-return indicator with an interaction between short interest and analyst coverage, predicting a negative interaction coefficient because our original positive relationship was expected to be stronger at lower coverage.

Exchange 2. Caught as wrong and corrected: data extraction (21 October 2026)

Prompt: (replying to Claude's note that "the memo's 5-day wording" was wrong) *"why is this wrong, professor mentioned that since short interest is announced every 2 weeks he wants us to change the data period into every 2 weeks calculated from daily data and convert to 2 weeks data. The 5 days return just means after the short interest announcement, does it increase future 5 days return/stock price?"*

Response (one sentence): Claude withdrew the claim, explaining that it had confused how often each stock is observed (already once per FINRA release, built from daily data) with how far ahead the return looks (the five-day horizon in our proposal), and in checking found that our regression script was using the two-week return as its main result.

What it changed: We knew because the professor's consultation and our proposal separate the two: the bi-weekly instruction sets how often the panel is observed, while the five-day return after each release is our research question. We made the five-day return the main result, as pre-registered, and kept the two-week release-to-release return as a robustness check instead of choosing whichever horizon looked more significant.

Exchange 3. Generated a Genuine Insight (23 October 2026)

Prompt: Our current research design and results suggest that higher short interest is associated with lower subsequent five-day returns. However, our findings cannot establish whether short covering is the underlying mechanism driving stock returns, as we originally hypothesized.

Response (one sentence): ChatGPT helped distinguish the predictive information contained in the level of short interest from the actual trading mechanism behind future price movements, using the informed-selling and short-covering literature.

What it changed: This helped us understand that our negative H1 coefficient does not make the research meaningless; instead, it provides evidence against our original positive prediction and is more consistent with the informed-selling interpretation. We consequently focused our discussion on what the regression can establish, while acknowledging that identifying short-covering pressure would require additional data on covering transactions and securities-lending conditions.